

Mani Vardhan Satha

+91-9959639020 | [E-Mail](#) | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Embedded Firmware Engineer with 2.5 years of experience on ARM-based systems, STM32, Raspberry Pi, and Qualcomm SoCs. Proficient in Embedded C, Python, and C++ for real-time systems using RTOS and Embedded Linux (OpenWRT, Yocto, Debian). Skilled in low-level protocols like I2C, SPI, UART, and BLE, as well as sensor integration and GUI development. Based in Hyderabad and open to relocating for onsite opportunities across India.

SKILLS

- **Programming Languages:** Embedded C, Python, C++, Shell Script
- **Microcontrollers:** STM32, ARM, ESP32, PIC, AVR, Raspberry Pi, Qualcomm SM6112
- **Protocols & Interfaces:** I2C, SPI, UART, MQTT, BLE, HTTP
- **Operating Systems:** Embedded Linux, Yocto, OpenWRT, Ubuntu, Debian, RTOS concepts
- **Object Detection:** OpenCV
- **Debugging Tools:** Oscilloscope (basic), Logic Analyzer (basic), UART/SPI/I2C protocol analyzers
- **Other Tools:** Git, JIRA, Bitbucket
- **Cloud & DevOps:** IoT Platforms, Sensor Integration, OpenCV, AWS

WORK EXPERIENCE

Caribou Technologies, Jul 2023–Present

Embedded Software Engineer, Hyderabad

- Engineered firmware for multiple IoT projects, employing Embedded C and managing version control with Git/SVN to ensure robust system integration.
- Implemented communication protocols including **I2C, SPI, UART, and MQTT** for efficient embedded device interactions.
- Optimized firmware to acquire and process real-time sensor data for CO₂, air quality, and pressure sensors, aligning with precise measurement instrumentation needs.
- Enhanced QC charging profile algorithms with performance-focused adjustments, leveraging strong analytical and coding skills.
- Worked on Qualcomm SM6112 SoC platform and Raspberry Pi, optimizing embedded Linux firmware (Debian 12) for wireless connectivity and real-time performance on GPU-less hardware.

Caribou Technologies, Dec 2022 - Jun 2023

Intern Hyderabad

- Developed embedded firmware for ARM-based platforms using Embedded C, contributing to improved system efficiency and reliability.
- Executed post-silicon validation techniques to test firmware and optimize hardware-software integration for embedded systems.
- Conducted comprehensive system-level testing to validate functionality, reliability, and performance of firmware implementations.

EDUCATION

Maturi Venkata Subba Rao, Hyderabad

BE, Electronics and Communication Engineering

- **Achievements:** Served as a senior coordinator in the National Service Scheme (NSS) and managed program operations for the "Sangamam" initiative.

PROJECTS

LIFETIME REGENERATIVE HYBRID VEHICLE

- Skilled in hybrid vehicle design and conversion, incorporating electric and gasoline systems for optimal performance.
- Experienced in retrofitting petrol engines to hybrid setups, employing advanced components such as hub motors and Li-ion batteries.

Wireless Solar Power Bank

- Designed and integrated hardware components, emphasizing portability and compatibility, to enable wireless charging of mobile devices.

CERTIFICATIONS

- AWS Machine Learning
- Developer Program—Accenture
- Python—Infosys